

acit1630_lesson_notes > week9 > normalization_lastnameInitial_firstname.db

SELECT * FROM Employee Query Editor Auto Reload Find Other Tools...

	emp	first	last
1	1001	Jane	Doe
2	1002	John	Doe
3	1003	Jane	Smith
4	1004	John	Smith
5	1005	Joe	Jones
+			

01 (a) normalization_FNF_employees

acit1630_lesson_notes > week9 > normalization_lastnameInitial_firstname.db

SELECT * FROM Children Query Editor Auto Reload Find Other Tools...

	emp REFERENCES Employee(emp)	c_first	c_birthday
1	1001 first:'Jane...	Mary	2001-01-02
2	1001 first:'Jane...	Sam	5/15/04
3	1002 first:'John...	Lisa	1/1/00
4	1002 first:'John...	David	5/15/01
5	1003 first:'Jane...	John	2010-05-04
6	1003 first:'Jane...	Pat	10/12/00
7	1003 first:'Jane...	Lee	2006-06-06
8	1003 first:'Jane...	Mary	8/21/04
9	1004 first:'John...	Michael	2007-04-06
10	1005 first:'Joe'...	Edward	10/21/05
11	1005 first:'Joe'...	Martha	10/15/99
+			

01 (b) normalization_FNF_children

⊗ NOT NULL constraint failed: orders.order_num Query: INSERT IN...

02_normalization_insertion_error1

⊗ NOT NULL constraint failed: orders.order_num Query: INSERT IN...

03_normalization_insertion_error2

normalization_lastnameInitial_firstname.db U X -- database: /Users/roger/Documents/1630 Untitled-1

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SELECT * FROM orders Schema Query Editor Auto Reload Find Other Tools... SQLite 3.51.2

	customer_num	first_name	last_name	street	city	state	zip	phone	order_num	order_date	item_num	
	INTEGER	TEXT	TEXT	TEXT	TEXT	TEXT	TEXT	TEXT	INTEGER NOT NULL PRIMARY KEY	datetime	INTEGER NOT NULL PRIMARY KEY	
1	1001	Jane	Doe	123 Elm St	Springfield	IL	62704	555-1234	2001	2024-03-01	5001	Lapt
2	1001	Jane	Doe	123 Elm St	Springfield	IL	62704	555-1234	2001	2024-03-01	5002	Wire
3	1002	John	Smith	456 Oak St	Chicago	IL	60616	555-5678	2002	2024-03-02	5003	Mech
4	1003	Alice	Brown	789 Pine St	Seattle	WA	98101	555-9012	2003	2024-03-03	5004	Gami
5	1004	David	Miller	567 Birch St	Los Angeles	CA	90001	555-7890	2004	2024-03-04	5005	Smar
+												

04_Orders_insertion

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SELECT * FROM orders Schema Query Editor Auto Reload Find Other Tools... SQLite 3.51.2

	customer _numb INTEGER 3	first_name e TEXT	last_name TEXT	street TEXT	city TEXT	state TEXT	zip TEXT	phone TEXT	order_num INTEGER NOT NULL PRIMARY KEY	order_date datetime	item_num INTEGER NOT NULL PRIMARY KEY	
1	1001	Jane	Doe	123 Elm St	Springfield	IL	62704	555-1234	2001	2024-03-01	5001	Lapt
2	1001	Jane	Doe	123 Elm St	Springfield	IL	62704	555-1234	2001	2024-03-01	5002	Wire
+												

05_normalization_deletion_anomaly

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SELECT * FROM orders Schema Query Editor Auto Reload Find Other Tools... SQLite 3.51.2

	customer _numb INTEGER 3	first_name e TEXT	last_name TEXT	street TEXT	city TEXT	state TEXT	zip TEXT	phone TEXT	order_num INTEGER NOT NULL PRIMARY KEY	order_date datetime	item_num INTEGER NOT NULL PRIMARY KEY	
1	1001	Jane	Doe	123 Elm street	Springfield	IL	62704	555-9876	2001	2024-03-01	5001	Lapt
2	1001	Jane	Doe	789 New Ave	Boston	MA	02108	555-9876	2001	2024-03-01	5002	Wire
+												

06_normalization_update_anomaly

normalization_lastnameInitial_firstname.db U X -- database: /Users/roger/Documents/1630 Untitled-1

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SELECT * FROM customer Schema Query Editor Auto Reload Find Other Tools... SQLite 3.51.2

	customer_num INTEGER PRIMARY KEY	first_name TEXT	last_name TEXT	street TEXT	city TEXT	state TEXT	zip TEXT	pho
1	1001	Jane	Doe	123 Elm St	Springfield	IL	62704	555-123
2	1002	John	Smith	456 Oak St	Chicago	IL	60616	555-567
3	1003	Alice	Brown	789 Pine St	Seattle	WA	98101	555-901
4	1004	David	Miller	567 Birch St	Los Angeles	CA	90001	555-789
+								

07(a)_normalization_2NF_Customers

normalization_lastnameInitial_firstname.db U X -- database: /Users/roger/Documents/1630 Untitled-1

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SELECT * FROM item Schema Query Editor Auto Reload Find Other Tools...

	item_num INTEGER PRIMARY KEY	title TEXT	price REAL	+
1	5001	Laptop	999.99	
2	5002	Wireless Mouse	29.99	
3	5003	Mechanical Keyboa...	89.99	
4	5004	Gaming Headset	79.99	
5	5005	SmartWatch	199.99	
+				

07(b)_normalization_2NF_Items

normalization_lastnameInitial_firstname.db U X -- database: /Users/roger/Documents/1630 Untitled-1

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SELECT * FROM orders Schema Query Editor Auto Reload Find Other Tools...

	order_num INTEGER PRIMARY KEY	customer_num INTEGER REFERENCES customer(customer_num)	order_date datetime	
1	2001	1001 firs...	2024-03-01	
2	2002	1002 firs...	2024-03-02	
3	2003	1003 firs...	2024-03-03	
4	2004	1004 firs...	2024-03-04	
+				

07(c)_normalization_2NF_Orders

normalization_lastnameInitial_firstname.db U X -- database: /Users/roger/Documents/1630 Untitled-1

acit1630_lesson_notes > week9 > normalization_lastnameInitial_firstname.db

SELECT * FROM order_items Schema Query Editor Auto Reload Find Other Tools...

	order_num INTEGER REFERENCES orders(order_num) PRIMARY KEY	item_num INTEGER REFERENCES item(item_num) PRIMARY KEY	has_shipped TEXT	
1	2001 customer_nu...	5001 title:'Lapt...	1	
2	2001 customer_nu...	5002 title:'Wire...	1	
3	2002 customer_nu...	5003 title:'Mech...	0	
4	2003 customer_nu...	5004 title:'Gami...	0	
5	2004 customer_nu...	5005 title:'Smar...	1	
+				

07(d)_normalization_2NF_OrderItems

normalization_lastnameInitial_firstname.db U X Normalization_Lab.docx U -- database: /Users/roger/

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SELECT * FROM **students** Schema Query Editor Auto Reload Find Other Tools...

	student_id <i>INTEGER PRIMARY KEY</i>	student_name <i>TEXT</i>	student_address <i>TEXT</i>	+
1	101	Alice Brown	123 Oak St	
2	102	Bob White	456 Pine St	
+				

08(a) – normalization_3NF_Students

normalization_lastnameInitial_firstname.db U X Normalization_Lab.docx U

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SELECT * FROM **instructors** Schema Query Editor Auto Reload F

	instructor_id <i>INTEGER PRIMARY KEY</i>	instructor_name <i>TEXT</i>	+
1	9001	Dr. Smith	
2	9002	Dr. Johnson	
+			

08(b) – normalization_3NF_Instructors

normalization_lastnameInitial_firstname.db U X Normalization_Lab.docx U -- database: /Users/roger/

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SELECT * FROM **courses** Schema Query Editor Auto Reload Find Other Tools...

	course_id <i>PRIMARY KEY</i>	course_name <i>TEXT</i>	instructor_id <i>INTEGER REFERENCES instructors(instructor_id)</i>	+
1	CSE101	Databases	9001	instructor_...
2	CSE102	Networking	9002	instructor_...
+				

08(c) – normalization_3NF_Courses

normalization_lastnameInitial_firstname.db U X Normalization_Lab.docx U -- database: /Users/roger/Documents/16

acit1630_lesson_notes > week9 > normalization_lastnameInitial_firstname.db

SELECT * FROM **enrollments** Schema Query Editor Auto Reload Find Other Tools...

	enrollment_id INTEGER PRIMARY KEY	student_id INTEGER REFERENCES students(student_id)	course_id TEXT REFERENCES courses(course_id)	+
1	1	101 student_name...	CSE101 course_na...	
2	2	102 student_name...	CSE102 course_na...	
3	3	101 student_name...	CSE102 course_na...	
+				

08(d) – normalization_3NF_Enrollmenments